Centre for Ocean Research

Applied Phycology Lab

The Applied Phycology Lab is equipped with advanced instrumentation such as a Continuous Flow Analyzer (AutoLab), ATR-FTIR, and a Multimode Reader to explore the multifaceted applications of algae in biotechnology, environmental sustainability, and industrial sectors. The lab primarily investigates algal adhesion and its ecological implications in marine environments, contributing to the development of eco-friendly strategies for combating marine biofouling. Additionally, it focuses on optimizing microalgal culture techniques to enhance biomass production, and supports spore-based seed production in macroalgae for ecological and commercial purposes.

Biomolecules Separation Lab

The Biomolecules Separation Lab houses advanced separation technologies, including Thin-Layer Chromatography (TLC) and High-Performance Liquid Chromatography (HPLC), for isolating novel molecules from natural sources. These systems allow for precise and efficient separation and quantitative analysis of biomolecules based on their interactions with stationary phases under high pressure. This capability supports the identification and development of new bioactive compounds with potential applications in environmental and biomedical fields.

Halophyte Conservation Lab

The Halophyte Conservation Lab undertakes applied research combining field surveys and laboratory experiments on seagrasses, seaweeds, and mangroves. The lab supports integrated strategies for the conservation and restoration of these highly productive coastal ecosystems. Facilities such as in-vitro micropropagation systems and mesocosm chambers are utilized for experiments on climate-induced stress, including warming and acidification. The lab also explores novel induction methods in halophyte cells for applications in natural product discovery, metabolic engineering, and conservation biology.



Livelihood and Skill Development Facility

Livelihood and Skill Development Facility is created to empower the youth with adequate skill sets that will enable their employment in relevant sectors and also improve productivity. The facility with associated labs and advanced instrumentation conducts capacity building through Skill development programmes. The trained youth will be encouraged with Institutional Seed fund to develop their own commercial products. The facility enables innovators to scale up their finished products into marketable and viable solutions through the support of Sathyabama Technology Business Incubator.

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